Notes: Acharya, Parlatore, & Sundaresan (RFS, 2025)
Financing Infrastructure in the Shadow of Expropriation

Contents

1 Big picture

- **Question.** Why do infrastructure projects remain underfinanced even when the social return to infrastructure appears high and global capital markets are large?
- **Core friction.** Infrastructure projects combine two agency problems. The private operator may shirk in developing, maintaining, or operating the project. The government may expropriate project rents after the project is built, for example by limiting user fees, changing regulation, retroactively taxing, terminating concessions, or reallocating ownership.
- **Public-private partnership structure.** The paper studies a project with three players: private financiers, a private operator, and the government. This is the natural setting for infrastructure because the government has coercive/regulatory power, the operator supplies effort, and outside financiers supply capital.
- **Double moral hazard.** The operator needs rents to exert high effort. But leaving rents to the operator makes government expropriation more tempting. If the government expropriates, operator incentives collapse and financiers anticipate lower expected repayment.
- **Main result.** The second-best contract uses **government guarantees to financiers against project failure**. These guarantees make the government internalize the downside of project failure, which gives it incentives not to expropriate operator rents.
- **Pecking order of fiscal tools.** When the double moral hazard is severe, the government uses fiscal capacity first for guarantees. As the project return rises or agency problems weaken, the contract adds government coinvestment. At still higher returns, financiers can receive a larger coupon/share of success payoffs.
- **Scale and feasibility.** Double moral hazard restricts both the extensive margin (some positive-NPV projects are not funded) and the intensive margin (funded projects are smaller than first best).
- **Developing versus developed economies.** Guarantees may be most valuable where expropriation risk is high, but those governments often have the least fiscal capacity. This helps explain why infrastructure gaps are especially persistent in developing economies.
- **Practice.** The model rationalizes features widely seen in infrastructure finance: guarantees, government coinvestment, tax subsidies, and development rights.

2 Model setup and economic environment

2.1 Players and stages

The model has three players: outside financiers, the government, and a private-sector operator. The project is implemented over four stages:

1. **Investment stage:** financiers and the government sign a financial contract and invest.
2. **Gestation stage:** an operator is appointed and the government signs an operational contract with the operator.
3. **Operating stage:** the operator chooses effort; the government may expropriate.
4. **Cash-flow stage:** the project payoff is realized and distributed.

Interpretation. This timeline makes the key commitment problem precise. Financing is raised before the government's temptation to expropriate is fully resolved.

2.2 Project technology and effort

The total scale of the project is

$$I \equiv I_f + I_g \leq \bar{I},$$

where I_f is private financing and I_g is government coinvestment. The project pays RI if successful and zero if it fails. If the operator exerts high effort, success probability is p_h; if she shirks, it is p_l, with

$$0 < p_l < p_h, \qquad \Delta p \equiv p_h - p_l, \qquad p_l R < 1 < p_h R.$$

If the operator shirks, she receives private benefits BI.

Interpretation. The project is socially worthwhile only with high effort. Therefore financing is not just about transferring money; it must preserve the operator's effort incentives.

2.3 Expropriation and user fees

The operational contract assigns the operator a success payoff $R_o I$. The government receives the residual project payoff after paying the operator and financiers. If the government expropriates the operator's rents, it receives an additional payoff CI, where C can be positive or negative.

Interpretation. A higher C means expropriation is more attractive to the government. Strong institutions, reputation, and legal enforcement can be interpreted as reducing C.

2.4 Financial contract

If the project succeeds, financiers receive $R_f I$, which can be interpreted as a success coupon. If the project fails, financiers receive $K_g I$ from the government, which is a guarantee. The government's success payoff is

$$R_g I = (R - R_f - R_o) I.$$

Interpretation. The paper's central distinction is that coupons and guarantees have opposite incentive effects. A higher coupon R_f reduces the government's success payoff and can worsen expropriation incentives. A higher guarantee K_g makes project failure costly for the government and can improve its incentives.

2.5 Fiscal capacity

The government has fiscal resources K_0 at the investment stage and K_1 at the cash-flow stage. It can use K_0 for direct coinvestment and can use remaining fiscal resources to honor guarantees.

Interpretation. Fiscal capacity is both a funding resource and an incentive device. The government uses scarce fiscal capacity to make its promises credible enough for financiers to participate.

3 Key constraints and equations

3.1 Operator incentive compatibility

The operator exerts high effort if

$$p_h R_o I \geq p_l R_o I + BI.$$

Equivalently,

$$R_o \geq A_o \equiv \frac{B}{\Delta p}.$$

Here A_o is the operator's agency rent.

Interpretation. The operator must keep at least A_o of the success return. If the government takes too much through expropriation, effort falls and the project becomes unattractive to financiers.

3.2 Government incentive compatibility

If the government agrees not to expropriate and induces high effort, it receives R_g with probability p_h and pays guarantees with probability $1 - p_h$. If it expropriates, the operator shirks and success probability falls to p_l. The government IC constraint is

$$p_h R_g - (1 - p_h) K_g \geq p_l (R - R_f + C) - (1 - p_l) K_g.$$

Using $R_o = A_o$, this can be written as

$$K_g + R_g \geq A_g, \qquad A_g \equiv \frac{p_l (A_o + C)}{\Delta p}.$$

Interpretation. A_g is the government's agency rent. It rises when the operator's agency rent rises. This is the double moral hazard: private moral hazard makes public moral hazard worse.

3.3 Financiers' participation

Financiers participate if expected payments cover the required return r on their investment:

$$r I_f \leq p_h R_f I + (1 - p_h) K_g I.$$

Interpretation. Financiers care about success payoffs and guarantees. A guarantee can relax participation while simultaneously improving the government's incentives.

3.4 Government participation

The government participates if its expected project payoff covers the return on its own investment:

$$\big[p_h R_g - (1 - p_h) K_g\big] I \geq r I_g.$$

Interpretation. The government cannot simply promise unlimited support; guarantees and coinvestment must be worthwhile from its own perspective.

3.5 Ability-to-pay constraints

The government's payments to financiers must respect fiscal resources:

$$R_f I \leq (R - A_o) I + K_0 + K_1 - I_g,$$

$$K_g I \leq K_0 + K_1 - I_g.$$

Interpretation. These constraints are the source of scale limits. If large guarantees are needed to solve incentives, the same guarantees restrict how large the project can be.

3.6 Planner's problem

The constrained-efficient contract maximizes the project surplus subject to the IC, IR, no-default, and scale constraints:

$$\max_{I_g \in [0, K_0],\ I_f \geq 0,\ K_g \geq 0,\ R_f \geq \underline{R}} (p_h R - r)(I_g + I_f),$$

subject to financiers' participation, government participation, government incentive compatibility, fiscal feasibility, and $I_g + I_f \leq \bar{I}$.

Interpretation. Because the objective is increasing in project scale when $p_h R > r$, all inefficiency comes from constraints: incentives, fiscal capacity, and limited commitment.

4 Main theoretical results

4.1 Benchmarks

- **Only operator moral hazard.** If the government has no incentive to expropriate, private moral hazard can make some projects infeasible, but once feasible the project is built at maximal scale.
- **Only government moral hazard.** If the operator has no moral hazard, government expropriation does not reduce effort. The project is funded whenever it is positive NPV and scale is maximal.

Interpretation. The central friction is not either moral hazard alone. The inefficiency comes from their interaction.

4.2 Inefficiency from double moral hazard

There are thresholds such that low net returns $R - A_o$ lead to no financing, intermediate net returns lead to financing at a constrained scale, and high net returns lead to maximal scale. The optimal scale is increasing in R and decreasing in A_o and A_g.

Interpretation. The model explains both why projects fail to attract funding and why funded projects may be too small. The same agency problem affects both extensive and intensive margins.

4.3 Government guarantees

Government guarantees are positive whenever the optimal financing contract is uniquely determined and the project does not naturally attain maximal scale. Guarantees decline as $R - A_o$ rises.

Interpretation. Guarantees are not a bailout add-on. They are the core incentive instrument that makes the government internalize the cost of project failure and commit not to expropriate.

4.4 Government coinvestment

Coinvestment becomes positive only when the project return is high enough relative to the double moral hazard. When the friction is severe, guarantees dominate coinvestment because using fiscal resources for coinvestment reduces resources available for guarantees.

Interpretation. The pecking order is: guarantees first, coinvestment later. This is the opposite of a simple funding-gap view in which the government should just add capital.

4.5 Return to financiers

The financiers' success payoff R_f rises above the minimum only when the project return is high enough and fiscal constraints become less binding.

Interpretation. Higher coupons can be useful for financiers' participation, but they reduce the government's success payoff and can worsen incentives. Hence they appear later in the pecking order.

4.6 Developed versus developing economies

A higher government benefit from expropriation C raises the required guarantee and lowers the optimal coupon to financiers. Higher fiscal resources increase project scale but do not change feasibility as long as resources are positive. Front-loaded fiscal resources are more efficient because they can support both coinvestment and guarantees.

Interpretation. Countries with weak institutions may need guarantees the most but may be least able to provide them. The model therefore produces an infrastructure-gap mechanism without relying on investor risk premia.

5 Extensions and mapping to practice

5.1 Costly state verification

The paper microfounds the financiers' minimum success repayment using a costly state verification problem. If financiers must verify success by forcing costly liquidation, the government optimally promises a minimum repayment to avoid verification.

Interpretation. This extension explains why the financial contract can look like debt: a coupon in success and a guarantee in failure.

5.2 Government expropriation from financiers

If the government can also expropriate from financiers, its limited commitment imposes a maximum credible payment. Greater default penalties or reputational costs increase the maximum scale but do not change basic feasibility.

Interpretation. Limited sovereign commitment creates an upper bound on project scale even when fiscal resources are large.

5.3 Expropriation in equilibrium

If operational contracts are enforced only with probability δ, the operator's agency rent becomes

$$\widehat{A}_o = \frac{B}{\delta \Delta p}.$$

Lower enforcement raises required operator compensation and can make more projects infeasible.

Interpretation. Weak contract enforcement tightens the extensive margin. Conditional on feasibility, the qualitative structure of the optimal contract remains the same.

5.4 Private-party coinvestment

If the operator can coinvest, guarantees should still be directed to financiers rather than the operator, because guarantees to the operator weaken effort incentives. The operator's coinvestment share is determined by her agency rent.

Interpretation. Operator skin in the game can arise endogenously, but insuring the operator against failure is counterproductive.

5.5 Externalities, development rights, and tax subsidies

Let D denote private-sector externalities such as development rights, and X denote government externalities such as political or fiscal gains from success. Development rights given to the operator reduce the operator's agency rent; rights given to financiers relax financiers' participation. Government externalities reduce the government's agency rent.

The paper shows that externalities increase both feasibility and scale. Development rights should go to the operator when the project payoff is low and operator/government incentives are most binding; they should go to financiers when the project payoff is high and financing constraints dominate.

Interpretation. This explains why infrastructure contracts bundle land-development rights, tax subsidies, and ancillary cash flows into the financing structure.

6 Identification and economic credibility

6.1 What identifies the theory mechanism

This is primarily a theory paper, so the strongest credibility comes from internal mechanism tests within the model:

- the two single-friction benchmarks do not generate the full inefficiency;
- combining operator and government moral hazard generates feasibility and scale restrictions;
- the optimal contract uses guarantees because guarantees improve government incentives, while coupons can worsen them;
- the pecking order of guarantees, coinvestment, and higher coupons maps to observed infrastructure contracts.

Interpretation. The paper's mechanism is disciplined by asking why specific real-world contract features are optimal, not merely common.

6.2 Empirical and institutional support

The discussion section provides supporting evidence rather than a causal empirical design. It points to disputes in Indian highway contracts, stressed Indian thermal power plants, and international examples of government expropriation risk. It also documents that government guarantees, coinvestment, development rights, and tax subsidies are prevalent in practice.

Interpretation. The evidence is meant to show that the model's frictions are realistic and that the contract features it predicts are observed in actual infrastructure finance.

6.3 What the paper cannot fully prove

- The India and global evidence are not causal tests of the model.
- The model abstracts from risk premia, political economy details, renegotiation bargaining, heterogeneous governments, and macro risk.
- Guarantees can create fiscal and distributional costs in practice; the model focuses on incentive efficiency.
- The analysis treats infrastructure finance through a stylized public-private partnership rather than a fully dynamic procurement process.

Interpretation. The paper is best read as a theory of why certain contracts solve incentive problems in infrastructure finance, not as a complete empirical measurement of infrastructure gaps.

7 Tables and figures

7.1 Figure 1 and Figure 2: Infrastructure need and model timeline

Read:

- Figure 1 shows that several U.S. infrastructure categories are close to, at, or above average life expectancy. Roads, dams, and water-treatment plants are especially salient examples.
- Figure 2 lays out the four-stage model: financial contracting and investment, operator appointment and operational contracting, effort and moral hazard, and final cash-flow distribution.

Economic message. The empirical motivation is that infrastructure needs are large, while the model explains why private capital may still not flow at sufficient scale.

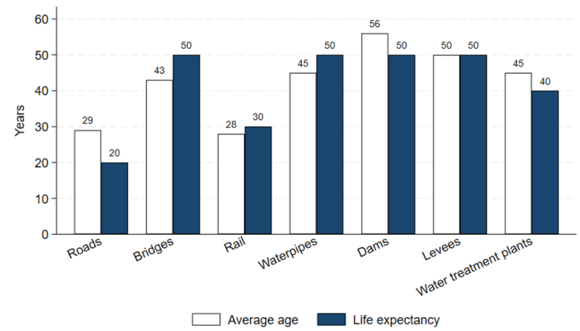

Figure 1
Average age and life expectancy of infrastructure in the United States as of 2022
Bar graph showing the average age and life expectancy of infrastructure in the United States as of 2002.
Source: American Society of Civil Engineers, https://infrastructurereportcard.org/ through Chinowsky (2021).

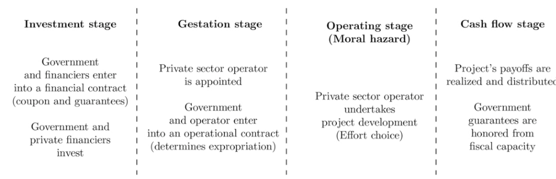

Figure 2
Timeline of the model
Diagram specifying the timeline of the model.

7.2 Figure 3 and Table 1: Payoff tree and equilibrium thresholds

Read:

- Figure 3 shows the success and failure payoffs. In success, financiers receive $R_f I$, the operator receives $R_o I$, and the government receives $(R - R_f - R_o)I$. In failure, financiers receive $K_g I$ and the government pays $K_g I$.
- Table 1 summarizes the return thresholds governing no financing, constrained financing, and maximal scale.

Economic message. The table and payoff tree show why guarantees are double-edged: they transfer resources to financiers in failure, but they also discipline the government.

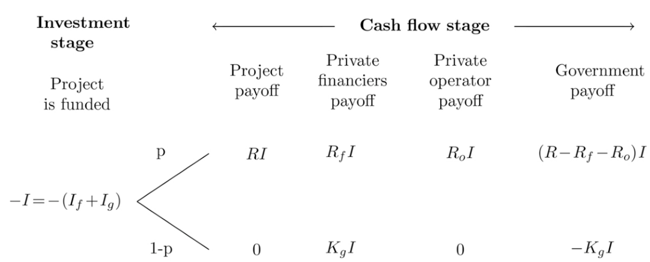

Figure 3
State space of possible outcomes and corresponding payoffs for various economic agents
Diagram depicting the possible outcomes of the model and the corresponding payoffs for the various economic agents.

Table 1
Equilibrium thresholds

$\underline{\Gamma}$	$\frac{r}{p_h}$
$\overline{\Gamma}$	$(1-p_h)A_g+\underline{R}$
Γ_I	$A_g-\frac{r-\underline{R}}{1-p_h}$
Γ_R	$A_g-\frac{(r-\underline{R})\overline{K}_1-\underline{R}r\overline{K}_0}{(r\overline{K}_0+(1-p_h)\overline{K}_1)}$
$\Gamma^\star$	$\frac{r}{p_h}+A_g-\frac{\overline{K}_1+r\overline{K}_0}{p_h\overline{I}}$

7.3 Figure 4: Pecking order of infrastructure finance

Read:

- Guarantees K_g are high when the project's return net of operator agency rent is low, then decline as $R - A_o$ rises.
- Government coinvestment I_g is initially zero, becomes positive only after the project is strong enough, and eventually reaches its fiscal maximum.
- Financiers' success return R_f stays at the minimum until the high-return region where their participation constraint requires more success-state compensation.

Economic message. The optimal contract has a fiscal pecking order: guarantees first, then coinvestment, then higher success-state repayment to financiers.

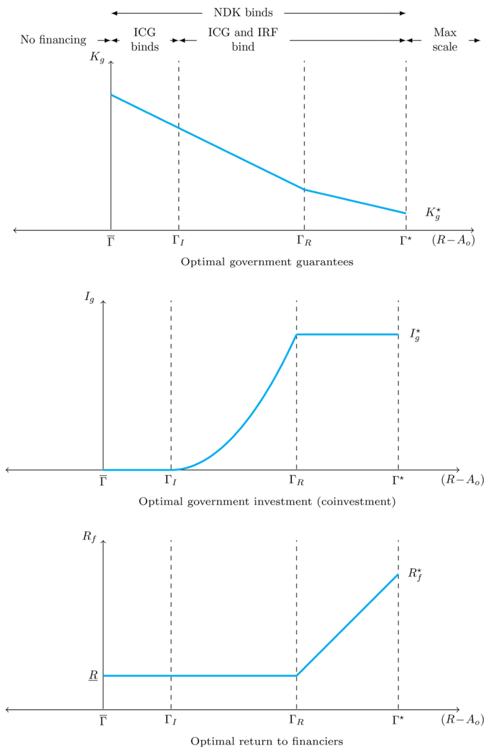

Figure 4
Optimal guarantee, government coinvestment, and return to financiers as a function of the project's return net of agency rents for the operator, $(R - A_o)$
Graphical representation of the government guarantee, government coinvestment, and return to financiers in the optimal contract as a function of the project return net of agency rents for the operator.

7.4 Tables 2 and 3: Highway disputes in India

Read:

- Table 2 shows that 66% of disputed Indian highway cases occur in the postaward/construction phase.
- Table 3 shows 382 litigation cases by driver: 260 arbitration-proceeds cases, 90 payment-related cases, and 32 wrongful-termination/debarment cases.
- Firms are petitioners in many payment and termination/debarment cases, consistent with concerns over government-side opportunism.

Economic message. The highway evidence supports the model's assumption that moral hazard is most important after the project is awarded, when the parties are locked into the relationship.

Table 2
Share of disputes by the phase of the contract life cycle for disputed highway projects in India from 2007 to 2020

Phase of the contract life cycle	Share (% of total)	Number of case
Preaward	2	11
Postaward / Construction	66	420
Postcompletion	0	1
Unclear/No data	32	203
Total	100	635

Source: Table reprinted from Mehta and Thomas (2022).

Table 3
Number of cases by drivers of litigation for disputed highway contracts in India from 2007 to 2020

Cause of dispute	Number of cases	Share (% of total)	NHAI as petitioner	Firm as petitioner
Arbitration proceeds related	260	68	123	137
Payments related	90	24	15	75
Wrongful termination/debarment	32	8	1	31
Total	382	100	139	243

Source: Table reprinted from Mehta and Thomas (2022).

7.5 Tables 4 and 5: Stressed thermal power plants in India

Read:

- Table 4 shows that 94% of the 34 stressed thermal power plants became stressed in the postaward/construction phase.
- Table 5 classifies 66 reasons for failure across the 34 projects. The paper attributes 22% of classifiable reasons to private moral hazard and 72% to public moral hazard; 6% are unclassified.
- The leading reasons include paucity of funds, land acquisition/handover delays, coal supply issues, and law-and-order problems.

Economic message. The power-plant evidence illustrates both sides of the double moral hazard, with public-sector failures especially prominent.

Table 4
Share of cases by the phase of the contract life-cycle for stranded or stressed thermal power plants in India initiated between 2007 and 2011

Phase of the contract life-cycle	Share (% of total)	Number of cases
Preaward	6	2
Postaward / Construction	94	32
Postcompletion	0	0
Unclear/No data	0	0
Total	100	34

This table categorizes the 34 stressed Indian thermal power plants by their phase of contract life-cycle, when the assets became stranded. The classification draws on the status of projects as delineated in the monthly Broad Status Reports by the Central Electricity Authority.

Table 5
Number of cases by drivers of fatal cause for stranded or stressed thermal power plants in India initiated between 2007 and 2011

Cause of failure	Number of cases	Share (% of total)	Private moral hazard	Public moral hazard	Unclassi-fied
Paucity of funds	17	26	NA	NA	17
Law and order problem	9	14	NA	NA	9
Legal issue	4	6	NA	NA	4
Delay of project status/clearances	3	5	0	3	0
Delay in acquisition/physical handover of land	11	17	0	10	1[a]
Coal supply issues	9	14	0	8	1[a]
Missing supply/transmission infrastructure	3	5	3	0	0
Operational issues/delays or other reasons	6	9	5	1[a]	0
Nonavailability of PPAs	4	6	0	4	0
Total	66 (34 total projects)	100	22%	72%	6%

The aggregate count of cases does not equate to the total of 34 stressed thermal power plants, as the analysis enumerates each reason for failure independently. This approach allows for a more nuanced understanding of the diverse factors leading to plant nonperformance. [a]See Appendix B1 for more details on these cases.

7.6 Figure 5: Global examples of government expropriation risk

Read:

- Figure 5 lists government expropriation risks across planning, operation, and termination phases.
- Examples include permit cancellation, tariff disputes, concession cancellation, contract breach, judicial risk, taxation risk, currency controls, and corruption/market distortion.
- The figure emphasizes that expropriation is broader than outright nationalization; it can operate through regulation, contracts, courts, taxes, and pricing rules.

Economic message. Government moral hazard is not limited to emerging markets or to one legal form. It appears as a broad class of actions that shift project rents away from private financiers and operators after sunk investment.

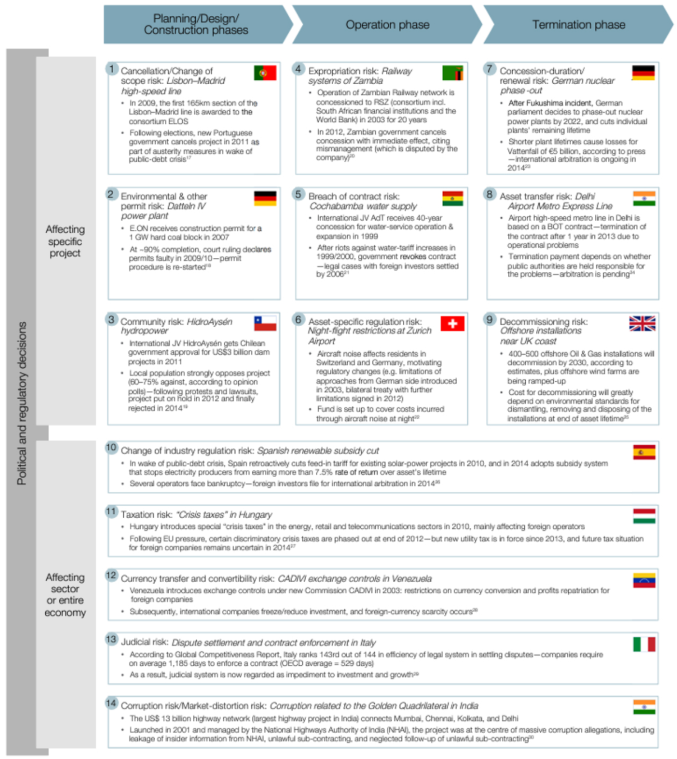

Figure 5
Examples of government expropriation affecting infrastructure projects
Diagram showing examples of government expropriation affecting infrastructure projects.
Source: World Economic Forum (2015, p.16).

8 Short summary

- Infrastructure finance is hard because private operators and governments both need incentives.
- Operator moral hazard requires leaving success rents to the operator.
- Leaving rents to the operator makes government expropriation more tempting.
- Government expropriation weakens operator effort and lowers expected repayment to financiers.
- Government guarantees to financiers solve part of this problem by making the government bear downside risk.
- Guarantees are most important when the project return is low relative to agency rents.
- Government coinvestment is useful only after the moral hazard problem is less severe.
- Higher success-state repayments to financiers appear later in the financing pecking order.
- Fiscal capacity and government commitment determine how large the project can be.
- Development rights and tax subsidies are useful because they can relax either operator incentives or financier participation.

9 Conclusion

9.1 Core conclusion

The paper provides a theory of infrastructure finance in which the main barrier is not simply a shortage of private capital. It is a double moral hazard problem. Operators need incentives to exert effort, while governments need incentives not to expropriate the rents that make operator effort possible.

9.2 Economic intuition

The key intuition is that government guarantees are not only insurance for financiers. They are an incentive device for the government. By committing fiscal resources in the failure state, the government makes project failure costly to itself. That gives it a reason to preserve operator rents, which improves effort, success probability, and ultimately financiers' willingness to invest.

9.3 Takeaway

Infrastructure finance works best when the contract aligns all three parties. Financiers need repayment, operators need rents, and governments need downside exposure. Guarantees, coinvestment, tax subsidies, and development rights are not arbitrary subsidies; in the model, they are instruments for making public-private infrastructure projects financeable under expropriation risk.